

Final Project :

Backdoor Attack on Federated Learning

Version: 1.2

Backdoor Attack on Federated Learning

- Write the code to apply the Backdoor Attack on Federated Learning (FL) system, especially for FL-based malware classification system (HW5, e.g., a modified version of HW5, based on binary content of malware, like HW3), and observe the results.

You can check the below paper and adopt some ideas:

1. [E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How To Backdoor Federated Learning," in Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics, vol. 108, S. Chiappa and R. Calandra, Eds., 26-28 Aug. 2020, pp. 2938-2948.](#)
 - [Source Code](#)
2. [H. Zhuang, M. Yu, H. Wang, Y. Hua, J. Li, and X. Yuan, "Backdoor Federated Learning by Poisoning Backdoor-Critical Layers," in *Proceedings of the International Conference on Learning Representations*, Messe Wien Exhibition and Congress Center, Vienna, Austria, May 7–11, 2024.](#)
 - [Source Code](#)
3. [V. Tolpegin, S. Truex, M. E. Gursoy, and L. Liu, "Data poisoning attacks against federated learning systems," in *Computer Security—ESORICS 2020: 25th European Symposium on Research in Computer Security, ESORICS 2020, Guildford, UK, September 14–18, 2020, Proceedings, Part I* 25, Springer, 2020, pp. 480–501.](#)
 - [Source Code](#)

- **Goal:**
 1. Misleading the FL-based malware classification system (your HW5)
 2. Keeping the functionality of malware: provide the proof, e.g., running Cuckoo Sandbox to get report

(Bonus) Provide a defense mechanism

- [Z. Zhang, X. Cao, J. Jia, and N. Z. Gong, "Fldetector: Defending Federated Learning Against Model Poisoning Attacks via Detecting Malicious Clients," in Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Aug. 2022, pp. 2545-2555.](#)
- [C. Xie, M. Chen, P.-Y. Chen, and B. Li, "Crfl: Certifiably robust federated learning against backdoor attacks," in *Proceedings of the International Conference on Machine Learning*, PMLR, 2021, pp. 11372–11382.](#)
- [Y.-C. Lai, J.-Y. Lin, Y.-D. Lin, R.-H. Hwang, P.-C. Lin, H.-K. Wu, and C.-K. Chen, "Two-phase defense against poisoning attacks on federated learning-based intrusion detection," *Computers & Security*, vol. 129, p. 103205, 2023.](#)
- [C. Xie, Y. Long, P.-Y. Chen, Q. Li, S. Koyejo, and B. Li, "Unraveling the connections between privacy and certified robustness in federated learning against poisoning attacks," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, 2023, pp. 1511–1525.](#)

Some Resources (Paper)

- [X. Gong, Y. Chen, Q. Wang, and W. Kong, "Backdoor attacks and defenses in federated learning: State-of-the-art, taxonomy, and future directions," *IEEE Wireless Communications*, vol. 30, no. 2, pp. 114–121, 2022.](#)
- [J. Lin, L. Dang, M. Rahouti, and K. Xiong, "ML attack models: Adversarial attacks and data poisoning attacks," *arXiv preprint arXiv:2112.02797*, 2021.](#)
- [L. Lyu, H. Yu, X. Ma, C. Chen, L. Sun, J. Zhao, Q. Yang, and S. Yu, "Privacy and robustness in federated learning: Attacks and defenses," *IEEE Transactions on Neural Networks and Learning Systems*, 2022.](#)
- [N. Rodríguez-Barroso, D. Jiménez-López, M. V. Luzón, F. Herrera, and E. Martínez-Cámara, "Survey on federated learning threats: Concepts, taxonomy on attacks and defences, experimental study and challenges," *Information Fusion*, vol. 90, pp. 148–173, 2023.](#)

Upload the result to Moodle

- Upload your code and the ***detailed*** reports to Moodle.
 - Code: You must add some comments in your code for easy understanding.
 - Report Files: Document (Word) and Presentation (PPT).
 - Including a table in your report outlining the responsibilities of each team member for this final project.
 - Zipped to a file. The file name should be “*Team Name_FinalProject.zip*”